

LHWAI Implementation Pack Index and Extraction Plan v0.1

Author: Ivan Kotov
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c Hardening Pack v0.1

c = a + b / L4 / SER / L4 Witness / ARL / Continuity Bundle

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Document class: implementation pack index / schema extraction plan / validator plan

Assertion class: C-A10 control-layer artifact

This document is a draft technical / architectural profile.

It is not legal advice.

It is not product certification.

It is not implementation-ready.

It is not deployment-ready.

No conformance-supported status is claimed unless separately validated by evidence, tests, and review.

LHWAI Implementation Pack Index and Extraction Plan v0.1

Schemas, fixtures, validator CLI, and conformance artifacts for Long-Horizon Witness Anchoring and Interpretation

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Status: Draft implementation-pack index v0.1

Layer: LHWAI / L4 Witness / ARL / Continuity Bundle / CLI Agent Mesh / GitHub-Zenodo release discipline

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0. Executive definition

This document turns the Long-Horizon Witness Anchoring and Interpretation Profile into implementation artifacts.

The normative profile defines:

```
witness epochs
digest ladder
anchor packets
external commitments
interpretation profiles
semantic freeze
challenge records
validation reports
red-line failures
conformance levels
```

This implementation pack defines the files needed to make those rules testable.

1. Package layout

Recommended layout:

```
docs/witness/long-horizon/
  Long_Horizon_Witness_Anchoring_and_Interpretation_Profile_v0_1.md
  LHWAI_Implementation_Pack_Index_and_Extraction_Plan_v0_1.md
schemas/
  long_horizon_witness_anchor_packet.schema.json
  witness_interpretation_profile.schema.json
  witness_interpretation_challenge_record.schema.json
  witness_anchor_validation_report.schema.json
  key_epoch_record.schema.json
fixtures/
  valid_anchor_packet_minimal.yaml
  valid_interpretation_profile_minimal.yaml
  valid_interpretation_challenge_policy_migration.yaml
  valid_validation_report_pass_with_gaps.yaml
  invalid_public_low_entropy_hash.yaml
  invalid_agent_self_exoneration.yaml
  invalid_missing_interpretation_profile.yaml
conformance/
  LHWAI_Test_Matrix_v0_1.md
  LHWAI_Red_Line_Failure_Cases_v0_1.md
tools/
  lhwai_validate.py
  lhwai_make_epoch_digest.py
  lhwai_make_anchor_packet.py
```

2. Required schemas

Schema	Purpose	Priority
long_horizon_witness_anchor_packet.schema.json	Validate anchor packet shape.	P0
witness_interpretation_profile.schema.json	Validate semantic decoder profile.	P0
witness_interpretation_challenge_record.schema.json	Validate reinterpretation/challenge records.	P0
witness_anchor_validation_report.schema.json	Validate later audit reports.	P0
key_epoch_record.schema.json	Validate key rotation and revocation epochs.	P1

3. Fixture requirements

3.1 Valid fixtures

Fixture	Must demonstrate
valid_anchor_packet_minimal.yaml	epoch digest + interpretation profile hash + packet hash
valid_interpretation_profile_minimal.yaml	schema/policy/reason-code registry hashes
valid_interpretation_challenge_policy_migration.yaml	append-only reinterpretation under new policy
valid_validation_report_pass_with_gaps.yaml	validation with declared missing/disputed events

3.2 Invalid fixtures

Fixture	Must fail because
invalid_public_low_entropy_hash.yaml	public anchor leaks guessable sensitive payload
invalid_agent_self_exoneration.yaml	agent validates its own contested high-risk event

Fixture	Must fail because
invalid_missing_interpretation_profile.yaml	anchor packet lacks required interpretation context
invalid_silent_reclassification.yaml	old event reclassified without challenge record
invalid_key_rotation_reset.yaml	key rotation treated as historical reset

4. Validator CLI requirements

A minimal validator CLI SHOULD support:

```
lhwai validate-anchor <anchor_packet>
lhwai validate-profile <interpretation_profile>
lhwai validate-challenge <challenge_record>
lhwai validate-report <validation_report>
lhwai validate-fixture-suite <fixtures_dir>
lhwai make-epoch-digest <event_manifest>
lhwai make-anchor-packet <epoch_digest> <profile>
```

The validator MUST check:

1. schema validity;
2. required hashes present;
3. interpretation profile references present;
4. privacy flags present;
5. no raw content in public packet;
6. low-entropy public hash risk field present;
7. no self-certifying compromised agent;
8. key epoch references where required;
9. gap status declared;
10. red-line failures.

5. Acceptance criteria

The implementation pack is minimally acceptable when:

```
all P0 schemas exist
all P0 fixtures exist
validator rejects all invalid fixtures
validator accepts valid fixtures
README explains local-only vs external commitment modes
conformance levels LHWAI-0..5 are machine-checkable at declaration level
```

6. Integration points

Area	Integration
L4 Runtime Enforcement	budget/hard-stop events enter witness epochs
Memory Drift Profile	memory promotion/compaction/quarantine emits witness refs

Area	Integration
Human Anchor Drift	anchor drift challenges create interpretation challenge records
90-Day Evidence Pack	monthly validation report and anchor packets included
GitHub/Zenodo release	redacted manifest / signed tag / DOI metadata used as external commitment
CLI Agent Mesh	package ledger and compromised-agent quarantine bind to anchor packets

7. Bridge set

Explicit bridge

LHWAI gives the long-horizon witness rules. This implementation pack makes them testable without redefining L4 Witness, ARL, or Continuity Bundle.

Quiet bridge I — cybernetics

A witness system cannot regulate historical drift if validation is manual folklore. The validator closes the loop by making the record checkable.

Quiet bridge II — information theory

A schema is a decoder contract. Without schemas and fixture failures, a digest is only a number. With schemas, it becomes verifiable structure.

Earth paragraph

A signed inspection note is useful. A binder with the note, drawing version, photo index, batch ticket, and later validation checklist is stronger. The implementation pack is that binder.

8. Open issues

ID	Issue	Priority
OI-LHWAI-IMPL-001	Write JSON schemas.	P0
OI-LHWAI-IMPL-002	Produce first fixture suite.	P0
OI-LHWAI-IMPL-003	Implement validator CLI.	P1
OI-LHWAI-IMPL-004	Add CI check for fixtures.	P1
OI-LHWAI-IMPL-005	Add release/DOI external commitment example.	P2